SAT Math Mastery · Session 4

Equivalent Expressions and Nonlinear Equations

Time 3 hours: Part A in class (about 2 hours 20 minutes of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.

Topics quadratic equations: discriminant, sum and product of solutions · absolute-value and rational equations · equivalent rational expressions · exponent rules and radicals · factoring and polynomial division · matching coefficients · undefined values

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

For nonzero real numbers p and q, the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) -14

(B) -2

(C) 2

(D) 14

2. HARD · DESMOS

The equation $x^2 - 9x + k = 0$, where k is a constant, has two real solutions that differ by 3. What is the value of k?

3. HARD · DESMOS

The equation $2x^2 - 10x + c = 0$, where c is a constant, has a product of solutions equal to 6. What is the sum of the reciprocals of the two solutions?

(A) $\frac{1}{5}$

(B) $\frac{1}{30}$

(C) $\frac{5}{6}$

(D) $\frac{6}{5}$

4. HARD · DESMOS

Let r and s be the two solutions of $x^2 - 7x + 3 = 0$. What is the value of $r^2 + s^2$?

5. HARD · DESMOS

The equation $x^2 - 5x + 6 = 0$ has solutions p and q. Which of the following equations has solutions $2p$ and $2q$?

(A) $x^2 - 10x + 24 = 0$

(B) $x^2 - 10x + 12 = 0$

(C) $x^2 - 5x + 12 = 0$

(D) $2x^2 - 5x + 6 = 0$

6. HARD · DESMOS

The equation $x^2 - bx + 27 = 0$, where b is a positive constant, has two real solutions, and one solution is 3 times the other. What is the value of b?

7. HARD · DESMOS

$$x^2 - 6x = 2|x - 3|$$

What is the sum of all real solutions of the equation above?

(A) 9

(B) 0

(C) 3

(D) 6

8. HARD · DESMOS

$$\frac{x}{x-4} = 2 + \frac{4}{x-4}$$

How many solutions does the equation above have?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

9. Hard · Desmos

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f(x)$ in the xy-plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x-intercept. What is the greatest possible value of a?

(A) $\frac{5}{2}$

(B) 9

(C) $\frac{9}{2}$

(D) $\frac{1}{2}$

10. Hard · Desmos

The expression $\frac{3}{x-4} - \frac{2}{x+1}$ is equivalent to which of the following, for $x \neq 4$ and $x \neq -1$?

(A) $\frac{x+11}{x^2+3x-4}$

(B) $\frac{x+11}{x^2-3x-4}$

(C) $\frac{x-11}{x^2-3x-4}$

(D) $\frac{5x-5}{x^2-3x-4}$

11. Hard · Desmos

The expression $\frac{5}{x+2} + \frac{3x}{x^2+x-2}$ is equivalent to which of the following, for $x \neq -2$ and $x \neq 1$?

(A) $\frac{8x-5}{x^2-x-2}$

(B) $\frac{3x+5}{x^2+x-2}$

(C) $\frac{8x+5}{x^2+x-2}$

(D) $\frac{8x-5}{x^2+x-2}$

12. HARD · DESMOS

The operation $\odot$ is defined for all real numbers $x \neq 1$ by $x^{\odot} = \dfrac{x+3}{x-1}$. Which of the following is equivalent to $x^{\odot} - 1$?

(A) $\dfrac{4}{x-1}$

(B) $\dfrac{2x+4}{x-1}$

(C) $\dfrac{2}{x-1}$

(D) $\dfrac{x+2}{x-1}$

13. HARD · DESMOS

If $a(x-2)(x+6) = a(x-h)^2 + k$ for all values of x, where $a \neq 0$, what is k in terms of a?

(A) $16a$

(B) $-16a$

(C) $-12a$

(D) $4a$

14. HARD · DESMOS

Which expression is equivalent to $x^4 - 2x^2 - 15$?

(A) $\left(x^2 - 15\right)\left(x^2 + 1\right)$

(B) $\left(x^2 - 1\right)\left(x^2 + 15\right)$

(C) $\left(x^2 - 5\right)\left(x^2 + 3\right)$

(D) $\left(x^2 + 5\right)\left(x^2 - 3\right)$

15. HARD · DESMOS

If $\dfrac{x^{a^2}}{x^{b^2}} = x^{20}$, $x > 1$, and $a + b = 4$, what is the value of $a - b$?

16. HARD · NO CALCULATOR NEEDED

If $x > 0$ and $\sqrt{x} = x\sqrt{x} = x^2 = \sqrt{x^2}$, what is the value of x?

17. HARD · DESMOS

If $s^2 = t^{-1}$ and $t^{1/3} = u^{-1/2}$, then what is the value of s in terms of u, for $s > 0$?

(A) $u^{3/4}$

(B) $u^{-3/4}$

(C) $u^{-1/2}$

(D) $u^{1/2}$

18. HARD · DESMOS

Let $\langle x \rangle$ be defined by $\langle x \rangle = \dfrac{3x+1}{x-2}$ for any x such that $x \neq 2$. Which of the following is equivalent to $\langle x \rangle - 3$?

(A) $\dfrac{6x-5}{x-2}$

(B) $\dfrac{7}{x-2} - 3$

(C) $\dfrac{3x-2}{x-2}$

(D) $\dfrac{7}{x-2}$

19. HARD · DESMOS

Let a and b be numbers such that $a^4 = b^3$. Which of the following is equivalent to $a\sqrt{b}$?

(A) $b^{4/3}$

(B) $b^{3/2}$

(C) $b^{5/4}$

(D) $b^{7/6}$

20. HARD · DESMOS

What is the value of $\left(\dfrac{3x}{y}\right)\left(\dfrac{x^{-1}}{(3y)^{-1}}\right)$?

(A) 6

(B) 9

(C) 1

(D) 3

21. HARD · NO CALCULATOR NEEDED

Find the value of $\dfrac{\sqrt{c+7}}{c} \cdot c\sqrt{c}$ if $c = 9$.

22. HARD · DESMOS

Evaluate $\left(\dfrac{5}{a^2} + \dfrac{4}{3a}\right)^a$ for $a = -3$.

23. HARD · DESMOS

For $x \neq -3$ and $x \neq 2$, the expression $\dfrac{x^2 - 2x - 15}{x^2 + x - 6}$ is equivalent to $\dfrac{x + a}{x + b}$, where a and b are constants. What is the value of $a - b$?

24. HARD · DESMOS

If $a^{\frac{3}{4}} = 5$ and $a > 0$, what is the value of $a^{-\frac{3}{2}}$?

25. Hard · Desmos

If $a^2 + b^2 = 4ab$, where $a \neq b$ and $b \neq 0$, what is the value of $\dfrac{a+b}{a-b}$?

(A) 2

(B) 3

(C) 6

(D) $\pm\sqrt{3}$

26. Hard · Desmos

The equation $\dfrac{ax+b}{x+2} = 3 - \dfrac{11}{x+2}$ is true for all values of x not equal to -2, where a and b are constants. What is the value of ab?

27. Hard · Desmos

The expression $10x^4 + 33x^2 + 20$ can be rewritten as $(5x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(5x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + d$?

28. HARD · DESMOS

$$\frac{40x+100}{2x^2-5x-25}+\frac{x^2y-5xy}{4xy}$$

Which of the following is equivalent to the expression above for $x>5$, $y>0$?

(A) $\dfrac{x^2-10x+105}{4(x-5)}$

(B) $\dfrac{x^2-10x+105}{4(2x+5)}$

(C) 5

(D) $\dfrac{x-1}{x+15}$

29. HARD · DESMOS

For real numbers p and q with $p^2+4q^2=72$, the product $(2x+p)(x+q)$ is equivalent to $2x^2+bx+c$, where b and c are constants. What is the greatest possible value of c?

30. HARD · DESMOS

For constants p, q and c, the equation

$$(2x-3)(x^2+px+q)=2x^3-5x^2+cx+9$$

is true for all real values of x. What is the value of c?

31. HARD · DESMOS

For a positive real number x, $\dfrac{\sqrt[3]{x}\cdot\sqrt[4]{x^3}}{\sqrt[6]{x}}=\sqrt[3]{k}$, where $k=2^{11}$. Which of the following is equal to x?

(A) $2^{\frac{44}{3}}$

(B) 2^4

(C) $2^{\frac{11}{3}}$

(D) $2^{\frac{121}{36}}$

32. HARD · DESMOS

For $x>0$ and any positive integer n, which of the following is equivalent to $\dfrac{\left(x^{2n+1}\right)^3+4x^{6n+3}-2x\left(x^{3n+1}\right)^2}{x^{6n+1}}$?

(A) $3x^4$

(B) $3x^{12n+4}$

(C) $3x^2$

(D) $5x^2$

33. HARD · DESMOS

For real numbers $k\neq 3$ and $x\neq 0$:

$$\frac{4(k-3)}{x(k-3)}+\frac{2}{x}-\frac{3}{4}=\frac{3}{8}$$

What is the value of x?

34. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $2187\left(\dfrac{27}{2}\right)^{x-2}$

(B) $2187\left(\dfrac{27}{2}\right)^{x+2}$

(C) $162\left(\dfrac{27}{2}\right)^{x-1}$

(D) $\dfrac{729}{4}\left(\dfrac{27}{2}\right)^{x-2}$

35. HARD · DESMOS

For integers a, b, c, and d, $(ax+b)(cx+d) = 6x^2 + kx + 5$ for all values of x, where k is a constant. Which of the following must be true?

I. k is odd
II. When k is divided by 4, the remainder is 1
III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{x^2-9}{x-3}$, for $x \neq 3$?

(A) x^2-3

(B) $x+9$

(C) $x+3$

(D) $x-3$

37. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{x^2+x-12}{x+4}$, for $x \neq -4$?

(A) x^2-3

(B) $x-3$

(C) $x+3$

(D) $x-4$

38. Medium · Desmos

If a and b are positive numbers such that $a^5 = b^2$, which of the following is equivalent to $b\sqrt{a}$?

(A) $b^{9/4}$

(B) $b^{1/5}$

(C) $b^{9/5}$

(D) $b^{6/5}$

39. Medium · Desmos

What is the quotient when $2x^3 - 3x^2 - 11x + 6$ is divided by $x - 3$ (the division has no remainder)?

(A) $2x^2 + 3x - 2$

(B) $2x^2 - 3x - 2$

(C) $2x^2 + 3x + 2$

(D) $2x^2 - 9x + 16$

40. Medium · No calculator needed

Which expression is equivalent to $4\,(2x + 7) - 5x$?

(A) $13x + 28$

(B) $3x + 28$

(C) $3x + 2$

(D) $13x + 7$

41. Medium · No calculator needed

Which expression is equivalent to $\left(5x^2 + 3x - 9\right) - \left(2x^2 - 7x + 4\right)$?

(A) $7x^2 - 4x - 5$

(B) $7x^2 + 10x - 5$

(C) $3x^2 + 10x - 13$

(D) $3x^2 - 4x - 13$

42. Medium · No calculator needed

An urban planner models the area of a rectangular section of a park as $3x^2 + ax$ square meters, where a is a constant. This expression is equivalent to $x\,(3x + 11)$. What is the value of a?

(A) 11

(B) 14

(C) 3

(D) 8

43. Medium · Desmos

The sum of $-3x^2 + 5x + 18$ and $6x^2 - 2x - 9$ can be written in the form $ax^2 + bx + c$, where a, b, and c are constants. What is the value of $a + b + c$?

44. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{2x^2 + 8x}{2x}$, for $x \neq 0$?

(A) $x + 8$

(B) $2x + 4$

(C) $x^2 + 4$

(D) $x + 4$

45. Medium · Desmos

If $x > 0$ and $\sqrt{x\sqrt{x}} = x^a$, what is the value of a?

46. Medium · Desmos

If $a^{b/5} = 32$ for positive integers a and b, where a is a prime number, what is the value of b?

47. Medium · Desmos

Which of the following is a factor of the polynomial $\left(x^2 - 9\right)(x + 7)$?

(A) $x + 5$

(B) $x - 2$

(C) $x + 3$

(D) $x - 4$

48. Medium · No calculator needed

If $k = m^2n^2$, k is not 0, and the values of m and n are both tripled, then the value of k increases by which of the following factors?

(A) 27

(B) 81

(C) 3

(D) 9

49. Medium · Desmos

In the equation $\sqrt{200} + \sqrt{200} = m\sqrt{n}$, m and n are integers such that $m > n$ and n is not divisible by the square of any integer greater than 1. What is the value of m?

50. Medium · Desmos

The value of y decreased by 5 is directly proportional to the value of x increased by 3. If $y = 7$ when $x = 1$, what is the value of x when $y = 19$?

51. Medium · Desmos

The expression $\dfrac{11p^2 + 11q^2}{9p^2 + 30p + 9}$ is undefined when

(A) $p = 0$ or $q = 0$

(B) $q = -\dfrac{1}{3}$ or $q = -3$

(C) $p = -3$ or $q = -\dfrac{1}{3}$

(D) $p = -\dfrac{1}{3}$ or $p = -3$

52. Medium · Desmos

For $x > 0$, which expression is equivalent to $\sqrt{(9x^2)^3 \cdot x^6}$?

(A) $27x^6$

(B) $27x^9$

(C) $81x^6$

(D) $3x^6$

53. Medium · Desmos

If $\left(\sqrt{p}\right)^6 = q^{-3}$ and $q = \dfrac{1}{4}$, what is the value of p?

54. MEDIUM · DESMOS

$\dfrac{\sqrt[3]{x^7}}{\sqrt{x}} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

55. MEDIUM · NO CALCULATOR NEEDED

If $x + \dfrac{1}{x} = 5$, what is the value of $x^2 + \dfrac{1}{x^2}$?

56. MEDIUM · DESMOS

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x \neq 2, -5$, where r and t are constants. What is the value of rt?

(A) 15

(B) 20

(C) 60

(D) -20

57. MEDIUM · DESMOS

$\dfrac{\left(\sqrt[3]{x^2}\right)^5}{x^2} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?